In the middle, we have a 500-step gradient descent procedure in which we iteratively compute the loss (which is the negative of our log likelihood) and *descend* on this loss based on its gradients. During this for loop, we keep track of the negative log likelihood values that we obtain, as well as the mean value adjusted in each step. This is so that after training, we could visually inspect these values to determine whether they have converged or not.

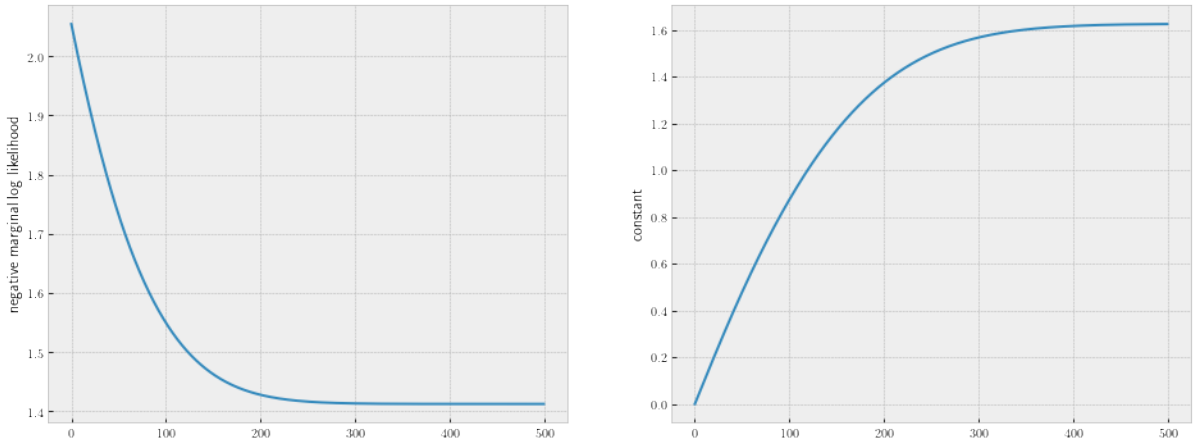


Figure 3.7 Running negative log likelihood (lower is better) and mean value during gradient descent. In both panels, the values have converged, indicating that we have arrived at an optimum.

Figure 3.7 includes this visualization, showing the running values for the negative log likelihood (which we aim to minimize) and the value for the mean function of the GP. Our loss consistently decreases (which is a good thing!) with increasing values of the mean constant, showing that a positive constant does give a higher likelihood than zero. Both curves plateau after 500 iterations, indicating that we have converged at an optimal value for the mean constant.

NOTE **The importance of monitoring the training process**

I recommend that you always plot out the progressive loss like we just did here when using gradient descent to see whether you have converged at an optimal value or not. Stopping before convergence might lead to poor performance for your model. While we won't be showing these progressive loss plots again throughout this chapter, the accompanying code does include them.

So far, we have learned how to use a zero mean function as the default for our GP model as well as to optimize the mean constant value with respect to the data likelihood. However, in many use cases, you might have prior knowledge about how the objective function is expected to behave, and thus prefer to incorporate more structure into your mean function.

For example, how can we implement the idea that the price of a house increases with a larger living area? Moving forward, we will learn to do this with a GP by using a linear or a quadratic mean function.

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